16 Compare different types Classification Algorithms and evaluate their performance.

Code:

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.naive_bayes import GaussianNB
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC
from sklearn.ensemble import RandomForestClassifier

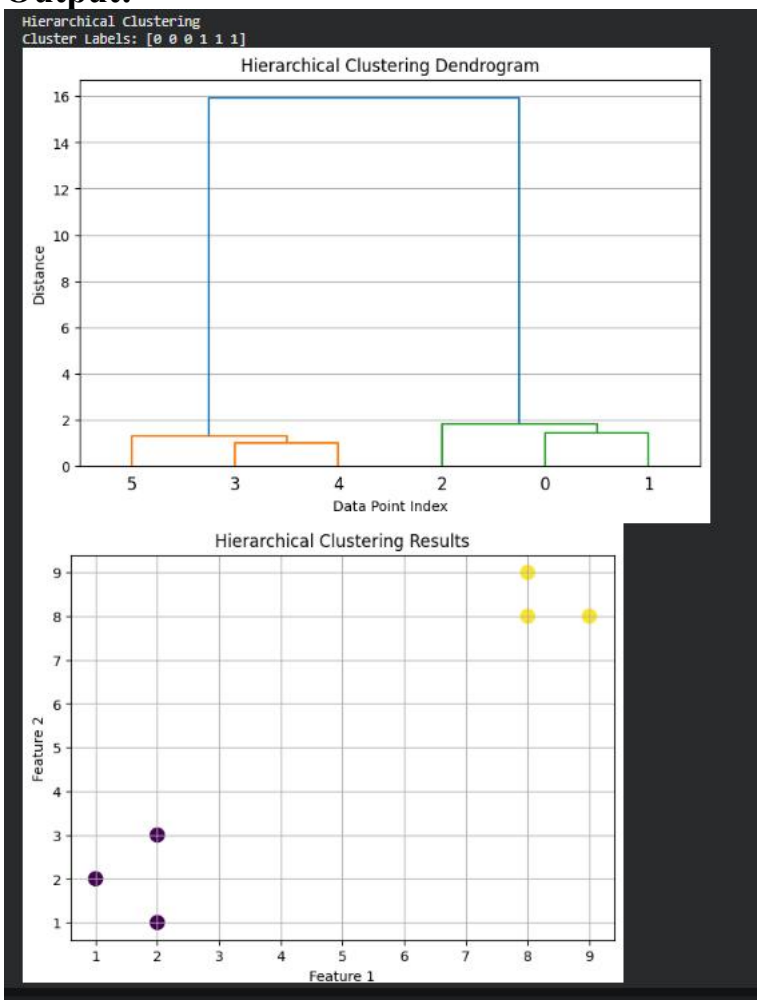
# Load dataset
iris = load_iris()
X = iris.data
y = iris.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

# Classification algorithms
models = {
    "KNN": KNeighborsClassifier(),
    "Decision Tree": DecisionTreeClassifier(random_state=42),
    "Naive Bayes": GaussianNB(),
    "Logistic Regression": LogisticRegression(max_iter=1000),
    "SVM": SVC(),
    "Random Forest": RandomForestClassifier(random_state=42)
}

# Compare performance
for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    accuracy = accuracy_score(y_test, y_pred)
  print(name, "Accuracy:", round(accuracy * 100, 2), "%")
```

Output:


Hierarchical Clustering
Cluster Labels: [0 0 0 1 1 1]
Hierarchical Clustering Dendrogram
Distance
Data Point Index
Hierarchical Clustering Results
Feature 2
Feature 1